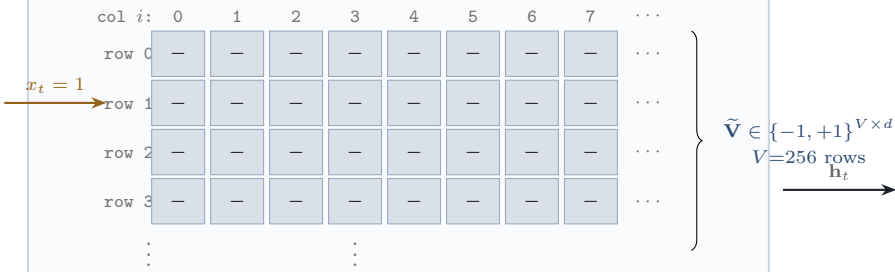


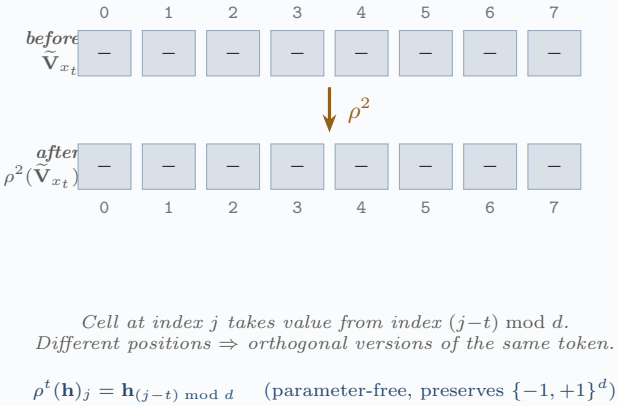
HDC-RWKV (nb12c) — Full Architecture Expansion

$V=256, d=256, L=1$, block size 16 • 16,434 deployment bytes • every learnable parameter bipolar at inference

Block 1. Token lookup — bipolar embedding table

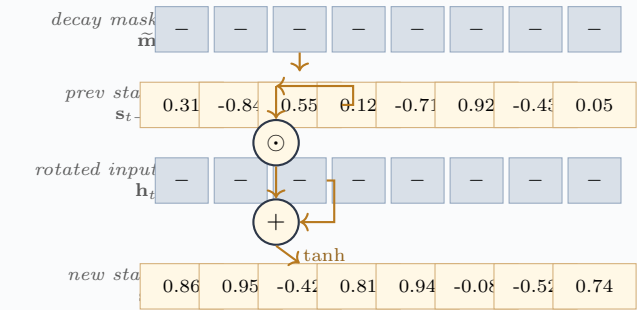


Block 2. Position binding ρ^t (cyclic right-rotation)

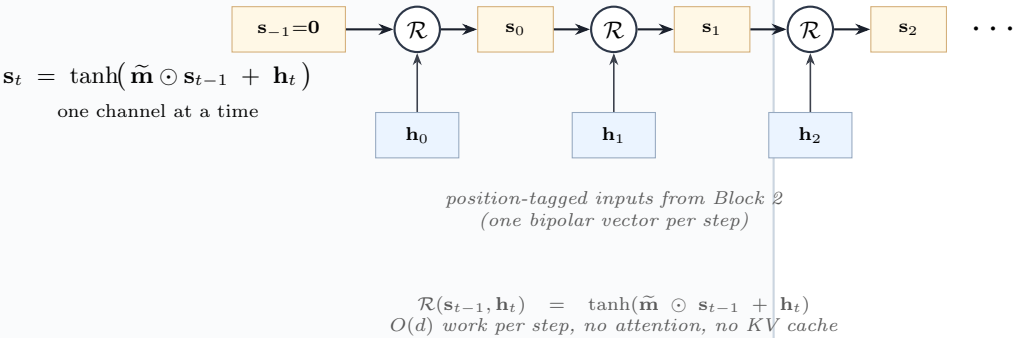


Block 3. Linear recurrence with continuous tanh state (one layer)

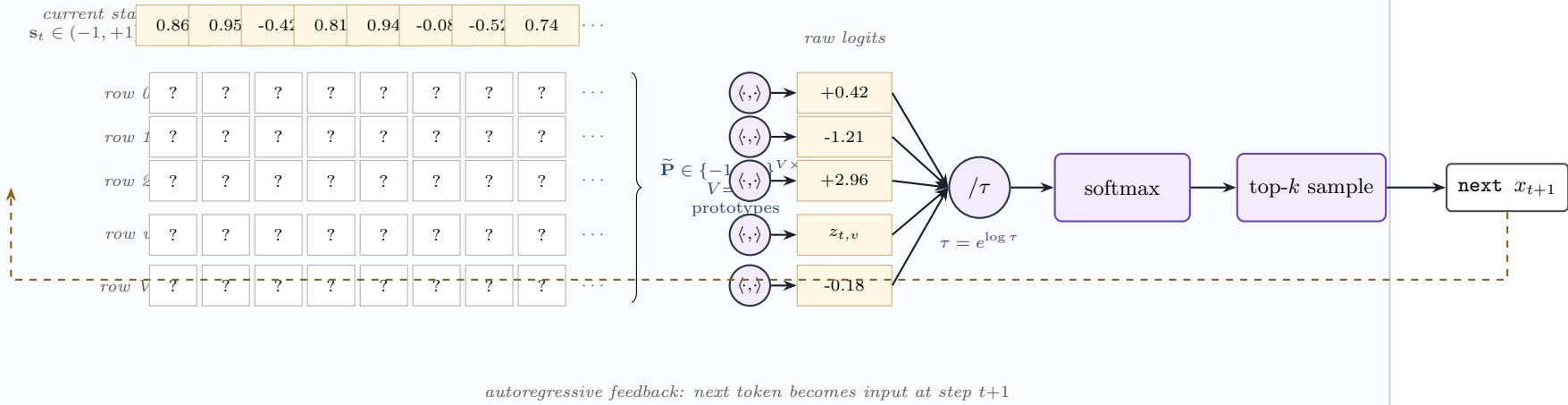
Per-channel detail at one timestep t



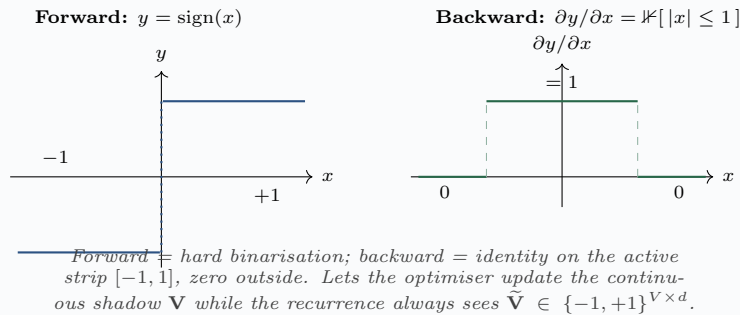
Unrolled across time



Block 4. Prototype-similarity output (one dot product per vocabulary item)



Aux A. Straight-Through Estimator ($\widetilde{\text{sign}}$)



Aux B. Deployment equivalent: XOR + popcount

Encode $-1 \mapsto 0, +1 \mapsto 1$. Then for $a, b \in \{-1, +1\}^d$:

$$\langle a, b \rangle = d - 2 \cdot \text{popcount}(\bar{a} \oplus \bar{b})$$

